

# Coincidence Classes in the ECDSA State Space: A Zero-Yield Theorem, with Exact Frequencies for Classes A, B and E

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## Abstract

We analyse a finite combinatorial object: the state space of an idealised ECDSA signer, together with a family of subsets defined by exact arithmetic coincidences between the *hidden* pair  $(s_{zk}, s_{rxk}) = (zk^{-1}, rxk^{-1})$  and quantities computable from public data. Three such subsets — the classes A, B and E of a previously circulated classification — are treated in closed form.

The organising result is a conservation law. Let  $R$  be *any* rule mapping the public data of a signature to a set  $C$  of candidate values for the hidden component. Then  $\Pr[s_{zk} \in C] = \mathbb{E}|C|/(n-2)$  exactly, so the success probability per tested candidate is the constant  $1/(n-2)$  for every rule, identical to sampling uniformly from  $\mathbb{Z}_n^* \setminus \{s\}$ . Classification cannot change the quality of candidates; it can change only their number. The classes are subsets, their frequencies are cardinalities, and nothing further is available.

Instantiating the theorem yields exact frequencies, proved and then confirmed by exhaustive enumeration of  $1.66 \times 10^9$  states. Class A emits  $\mathbb{E}|C| = \Pr[Q] \rightarrow 1/4$  candidates per signature; class E emits  $5/6$  and is shown to be *exactly* the affine locus  $s_{zk} = 3s_{rxk} + 2$ , whence  $\Pr[E] = 2[(n-3)/3]/(n-1)^2 \sim 2/(3n)$ ; class B emits  $\Theta(n^{-1} \log n)$ , and its multiplier satisfies an *integer* quadratic, so no modular square root arises. Three structural facts follow: the qualification rate is exactly  $1/4$  on every curve, being the area of a lattice region; the “Goldbach” offset distinguishing class E is one of  $\Theta(n)$  interchangeable choices; and the recovery formulas of the classification are a single Möbius bijection in disguise.

On secp256k1 the frequencies are  $2^{-258.00}$ ,  $2^{-256.58}$  and  $\approx 2^{-506.9}$  per signature. We report them because the question is natural, not because they are decisive: by the main theorem the entire framework improves on uniform random key guessing by a factor of exactly  $1 + 1/(n-2) \approx 1 + 10^{-77}$ , and that residual advantage is bought by the observation  $s_{zk} \neq s$  rather than by any of the algebra. This is a negative result. Its value is that it closes a natural line of enquiry with proofs rather than with estimates.

**Keywords:** ECDSA · secp256k1 · lattice-point counting · negative result · hidden number problem · nonce distribution

**MSC 2020:** 11T71, 94A60, 11H06, 05A15

**Statement of non-disclosure.** This paper reports no vulnerability. It describes no attack, no partial attack and no weakening of any deployed parameter set. Its main theorem proves that a family of proposed techniques cannot outperform exhaustive key search, which is itself  $\approx 2^{128}$  times more expensive than the generic attack the curve was designed to resist. Readers arriving from the earlier “vulnerability” framing of this project’s own preprints are directed to Section 11, in which that framing is withdrawn.

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# 1 Introduction

## 1.1 Motivation

ECDSA signing computes  $s = (z + rx)k^{-1} \bmod n$ , and the scalar  $s$  splits additively into a message part and a key part,

$$s_{zk} = zk^{-1} \bmod n, \quad s_{rxk} = rxk^{-1} \bmod n, \quad s \equiv s_{zk} + s_{rxk} \pmod{n}.$$

Both summands are secret. An analyst with access to  $(z, r, s)$  can nevertheless form auxiliary integers — most naturally  $s_{zr} = zr \bmod n$ , the difference  $d = s_{zr} - s$ , and the remainder  $a = s \bmod d$  — and ask whether the hidden summands happen to stand in an exact arithmetic relation to them. When they do, and when the relation pins  $s_{zk}$  to a small set of values, the private key follows immediately, because  $s_{zk}$  determines the nonce and the nonce determines the key.

Such coincidences do occur. On reduced-size test curves they occur often enough to be catalogued, and a previously circulated framework for this project catalogued five of them, labelled A through E, deriving closed-form key-recovery procedures for A, B and E. Empirically the classes thin out as the curve order  $n$  grows, and the natural questions are how fast, how far, and whether the thinning is a barrier or merely a nuisance.

This paper answers all three, and then shows that answering them was unnecessary.

## 1.2 Contributions

1. **A conservation law (Theorem 7).** For every *abscissa-oblivious* rule — one that treats  $r$  as an opaque scalar and never invokes the group law — the probability of a correct guess of  $s_{zk}$  equals the expected number of candidates emitted, divided by  $n - 2$ . The yield per candidate is a constant, independent of the rule, so no such classification can beat uniform random guessing (Corollary 8); a rule that does beat it is obliged to solve ECDLP (Corollary 9). We claim no novelty for the underlying fact, which any cryptographer would regard as a restatement of what it means for a signature scheme to be correctly randomised. What the theorem contributes is a boundary: it converts an open-ended hunt for further coincidence classes into a closed question, and it makes precise which property a genuine improvement would have to violate.
2. **A structure theorem for class E (Theorem 25).** The three-clause “Goldbach-type” definition of class E is equivalent to the single affine condition  $s_{zk} = 3s_{rxk} + 2$  over  $\mathbb{Z}$ . Its exact cardinality follows (Theorem 27), as does the fact that the distinguished offset is arbitrary (Theorem 29).
3. **Exact frequencies where the prior work had heuristics.**  $\Pr[\mathcal{Q}] = 1/4 + O(n^{-1} \log n)$  (Theorem 14);  $\Pr[A] = \Pr[\mathcal{Q}]/(n - 2)$ , an identity (Theorem 18);  $\Pr[E] = 2\lfloor(n - 3)/3\rfloor/(n - 1)^2$ , an identity (Theorem 27);  $\Pr[B] = \Theta(n^{-2} \log n)$  (Proposition 22).
4. **Class B lives over  $\mathbb{Z}$  (Theorem 20).** Its multiplier satisfies an integer quadratic whose discriminant must be a perfect square — a far stronger constraint than quadratic residuosity modulo  $n$ , and one that removes the modular square root from the published procedure. Wrap-around instances are shown to require  $m \leq 2$  together with an exact auxiliary relation (Lemma 21).
5. **Verification.** Exhaustive enumeration of the full state space for seven primes ( $1.66 \times 10^9$  states), Monte-Carlo runs at  $n \in \{9967, 99667\}$ , and comparison against three independently generated test-curve datasets spanning five orders of magnitude in  $n$ . Every identity is confirmed to the precision of the enumeration; every asymptotic estimate is confirmed to within Poisson fluctuation.
6. **Errata (Section 11).** Three corrections to previously circulated material, one of them to an earlier draft of this paper. The most consequential is that the published class-E rate  $1/(n - 1)$  overstates the truth by 50% and was already excluded at  $32\sigma$  by the data accompanying it.

## 1.3 What is not a contribution

Honesty about weight is part of the exercise. The following are *not* contributions and are not presented as such. The recovery formulas: Proposition 3 shows they are one Möbius bijection. The taxonomy A–E itself: an arbitrary partition of the analyst’s bookkeeping rather than of the object. The observation that rare coincidences occur at rate  $\Theta(1/n)$ : folklore. And, as noted above, the substance of Theorem 7,

which is a formalisation rather than a discovery. The weight of this work is that of a technical note recording a closed line of enquiry, and it is circulated as such.

## 1.4 What this paper does not claim

It is not a vulnerability disclosure. It contains formulas that map a *correct guess* of  $s_{zk}$  to the private key, and those formulas are correct; but by Proposition 3 they constitute a single Möbius bijection, that is, a change of coordinates on a search space of size  $n - 1$ , and by Theorem 7 no public-data rule can make the guess better than uniform. Reading them as an attack is a category error.

Nor, and this must be stated carefully, is the obstruction information-theoretic. ECDSA has no information-theoretic security whatever:  $r$  is the abscissa of  $R = kG$ , so it determines  $R$  up to the two points  $\pm R$  (and, on secp256k1, an  $x$ -coordinate  $r + n$  occurring with probability  $\approx 2^{-128}$ ), hence  $k$  up to a handful of values, hence  $x$ . The conditional entropy of the private key given one signature is a couple of bits, not  $\log_2 n$ . What makes ECDSA hard is that extracting  $k$  from  $r$  is an instance of the elliptic-curve discrete logarithm problem. Section 4 therefore proves a statement about a restricted class of algorithms — those that treat  $r$  as an opaque scalar and never invoke the group law — and Corollary 9 makes the boundary explicit: any rule that beats the bound is obliged to solve ECDLP.

## 1.5 Related work

Attacks that do recover ECDSA keys exploit a defect in the signer rather than a coincidence in the output. Repeated nonces yield the key from two signatures by elementary algebra and have been found repeatedly in the wild [3, 4]. Partially known or biased nonces reduce to the Hidden Number Problem of Boneh and Venkatesan [2] and are solved by lattice reduction [7, 10, 11]; a handful of leaked bits over a few dozen to a few million signatures suffices, as demonstrated against real deployments by Breitner and Heninger [4], the Minerva group [8], and LadderLeak [1], the last operating on less than one bit of leakage per signature. Deterministic nonce derivation [14] removes the randomness failure mode at the cost of introducing a fault-attack surface. The common feature is that each exploits  $\Theta(\ell)$  bits of information that is accessible by cheap algebra, whereas the classes studied here supply none that an abscissa-oblivious rule can reach (Theorem 7). They are therefore not a weaker point on the same spectrum but a different category. The information that does remain in a signature —  $r$  determines  $k$  up to sign — is reachable only by solving ECDLP, which is the province of [13] and not of any classification.

Generic attacks on the underlying discrete logarithm cost  $\Theta(\sqrt{n}) \approx 2^{128}$  operations on secp256k1 [13]; the curve parameters are those of SEC 2 [5], and the signature scheme is standardised in FIPS 186-5 [12]. The counting arguments below use Dirichlet’s divisor theorem in the form given by Hardy and Wright [6], and Landauer’s principle [9] is quoted once, for scale only.

## 1.6 Organisation

Section 2 fixes notation and proves the master identity. Section 3 states the probability model and its structure lemma. Section 4 proves the main theorem; a reader interested only in the conclusion may stop there. Sections 5–8 instantiate it for the qualification event and for classes A, B, E. Section 9 reports the verification, Section 10 evaluates on secp256k1, Section 11 records the errata, and Section 12 discusses falsifiability and limitations.

# 2 Preliminaries

## 2.1 Signatures and hidden components

Let  $E/\mathbb{F}_p$  be an elliptic curve with base point  $G$  of prime order  $n$ , and let  $Q = xG$  be the public key of  $x \in \{1, \dots, n - 1\}$ . For a message hash  $z$  and nonce  $k$ , both in  $\{1, \dots, n - 1\}$ ,

$$r = [kG]_x \bmod n, \quad s = (z + rx)k^{-1} \bmod n. \quad (1)$$

We write  $\mathbb{Z}_n^* = \{1, \dots, n - 1\}$  and identify every element of  $\mathbb{Z}_n$  with its representative in  $\{0, \dots, n - 1\}$ . This identification is essential, because the conditions below mix modular arithmetic with ordinary integer comparison, divisibility and order. Relations meant over  $\mathbb{Z}$  rather than over  $\mathbb{Z}_n$  are marked as such.

**Definition 1** (Hidden components).  $\zeta := s_{zk} = zk^{-1} \bmod n$  and  $\xi := s_{rxk} = rxk^{-1} \bmod n$ , so that  $s \equiv \zeta + \xi \pmod{n}$ . Neither is observable from  $(z, r, s)$ .

Since  $\zeta, \xi \in \{1, \dots, n-1\}$ , the integer sum  $\sigma := \zeta + \xi$  lies in  $\{2, \dots, 2n-2\}$  and satisfies exactly one of

$$\sigma = s \quad (\text{no wrap-around}), \quad \text{or} \quad \sigma = s + n \quad (\text{wrap-around}). \quad (2)$$

The distinction is invisible publicly and matters for classes B and E.

**Definition 2** (Public auxiliaries).  $s_{zr} := zr \bmod n$ ; when  $s_{zr} > s$ ,  $d := s_{zr} - s$  and  $a := s \bmod d$ , the last an integer remainder. All are computable from the signature and the signed message.

## 2.2 The master identity

**Proposition 3** (Master identity; recovery is a relabelling). (i) If  $\hat{s} = \zeta$ , then  $x = z(s - \hat{s})(r\hat{s})^{-1} \bmod n$ .

(ii) The map  $\varphi : \mathbb{Z}_n^* \setminus \{s\} \rightarrow \mathbb{Z}_n^*$ ,  $\varphi(\hat{s}) = z(s - \hat{s})(r\hat{s})^{-1} \bmod n$ , is injective. Hence knowing  $\zeta$  is equivalent to knowing  $x$ , and the formula in (i) transports no information from public data to the secret.

*Proof.* (i) From Definition 1,  $k = z\zeta^{-1}$  and  $\xi = (s - \zeta) \bmod n$ . Substituting into  $\xi = rxk^{-1}$  gives  $x = \xi kr^{-1} = (s - \zeta) \cdot z\zeta^{-1} \cdot r^{-1} = z(s - \zeta)(r\zeta)^{-1}$ . (ii) Expand:  $\varphi(\hat{s}) = zsr^{-1}\hat{s}^{-1} - zr^{-1}$ . Inversion is a bijection of  $\mathbb{Z}_n^*$ , and  $t \mapsto (zsr^{-1})t - zr^{-1}$  is an affine bijection since  $zsr^{-1} \neq 0$  in the field  $\mathbb{Z}_n$ . A composition of bijections is injective.  $\square$

Proposition 3(ii) governs the subject. Searching for  $x$  through  $\hat{s}$  is a change of coordinates on a space of size  $n - 1$ . Every class below is therefore no more than a rule for proposing candidate values of  $\zeta$ , and only two things about such a rule matter: how often the proposal is correct, and how many proposals it costs.

## 3 The probability model

**Assumption A** (Ideal signer).  $z, x$  and  $k$  are drawn independently and uniformly from  $\mathbb{Z}_n^*$ .

**Assumption B** (Abscissa heuristic).  $r = [kG]_x \bmod n$  is modelled as uniform on  $\mathbb{Z}_n^*$  and independent of  $(z, x, k)$ .

Assumption A is the definition of a correctly implemented signer; when it fails, ECDSA is broken by the mechanisms of Section 1.4, which have nothing to do with this paper. Assumption B is a genuine heuristic, since  $r$  is a deterministic function of  $k$ . It is adopted only to obtain closed forms. Every prediction derived from it is checked in Section 9 against datasets generated on actual elliptic curves, where no such assumption was made; agreement is within Poisson fluctuation across five orders of magnitude in  $n$  and eight in sample size. Assumption B is therefore not load-bearing for the numerical conclusions.

**Lemma 4** (Structure of the state). Put  $u := kr \bmod n$ . Under Assumptions A and B:

(i)  $(\zeta, \xi, u)$  is uniform on  $(\mathbb{Z}_n^*)^3$ ;

(ii) the map  $(\zeta, \xi, u) \mapsto (\zeta, s, s_{zr})$  with  $s = \zeta + \xi$  and  $s_{zr} = \zeta u$  (both mod  $n$ ) is a bijection onto

$$\mathcal{S} = \{(\zeta, s, s_{zr}) : \zeta \in \mathbb{Z}_n^*, s \in \mathbb{Z}_n \setminus \{\zeta\}, s_{zr} \in \mathbb{Z}_n^*\}, \quad |\mathcal{S}| = (n-1)^3,$$

so  $(\zeta, s, s_{zr})$  is uniform on  $\mathcal{S}$ ;

(iii) consequently  $s_{zr}$  is uniform on  $\mathbb{Z}_n^*$  and independent of  $(\zeta, s)$ , and for each fixed  $s \neq 0$  the conditional law of  $\zeta$  given  $(s, s_{zr})$  is uniform on  $\mathbb{Z}_n^* \setminus \{s\}$ , a set of exactly  $n - 2$  elements.

*Proof.* (i) Fix  $k, r$ . Then  $\zeta = zk^{-1}$  is a bijective image of the uniform  $z$ , hence uniform on  $\mathbb{Z}_n^*$ ;  $\xi = rxk^{-1}$  is a bijective image of the uniform  $x$ , hence uniform; they are independent because  $z \perp x$ . Neither depends on  $u = kr$ , a function of  $(k, r)$  alone; by Assumption B,  $u$  is uniform on  $\mathbb{Z}_n^*$  and independent of  $(z, x)$ , hence of  $(\zeta, \xi)$ . (ii) The map is invertible by  $\xi = s - \zeta$ ,  $u = s_{zr}\zeta^{-1}$ , and its image is exactly  $\mathcal{S}$ , since  $\xi \neq 0 \iff s \neq \zeta$  and  $u \neq 0 \iff s_{zr} \neq 0$ . A bijection carries the uniform law to the uniform law, and  $|\mathcal{S}| = (n-1)^3$  because for each  $\zeta$  there are  $n-1$  admissible values of  $s$ . (iii) Immediate from the product structure of  $\mathcal{S}$ .  $\square$

Lemma 4(iii) is the workhorse. Within the model, after the analyst has computed everything computable by modular arithmetic on  $(z, r, s) \rightarrow s_{zr}, d, a$ , and any function of them — the hidden component  $\zeta$  remains uniform over  $n - 2$  possibilities.

The reader should be clear about the force of this. It is a statement about the model, not about ECDSA. Assumption B is what makes it true, and Assumption B is false:  $r$  determines  $R = kG$  up to sign, hence  $k$  up to a handful of candidates, hence  $\zeta$  and  $x$ . The residual entropy of  $\zeta$  given a real signature is one or two bits. What Assumption B does is sever precisely the link — from the abscissa back to the scalar — whose difficulty is the entire security of the scheme. That severance is legitimate here for one narrow reason, made precise in Definition 6: the classes under study never touch the curve. It would be illegitimate for any conclusion about ECDSA in general, and no such conclusion is drawn.

## 4 The zero-yield theorem

**Definition 5** (Public-data rule). A rule is a map  $R$  from the public data of a signature —  $z, r, s$ , and anything computable from them, such as  $s_{zr}, d, a$  — to a finite candidate set  $C = R(z, r, s) \subseteq \mathbb{Z}_n^*$  for the hidden component  $\zeta$ . Randomised rules are permitted, provided their randomness is independent of the signer's. The rule *succeeds* if  $\zeta \in C$ .

**Definition 6** (Abscissa-oblivious rule). A rule is *abscissa-oblivious* if its output distribution is unchanged when the signer's  $r = [kG]_x \bmod n$  is replaced by an independent uniform draw from  $\mathbb{Z}_n^*$ . Equivalently: the rule treats  $r$  as an opaque element of  $\mathbb{Z}_n$  and never invokes the group law of  $E$  — it never forms a curve point, never adds or doubles, never solves for a discrete logarithm.

Classes A, B and E are abscissa-oblivious: every quantity they compute ( $s_{zr} = zr, d, a, m_1, m_2, \alpha, \beta$ ) is obtained by ring operations on  $z, r, s$  in  $\mathbb{Z}_n$  together with integer division. The single scalar multiplication each performs is a *verification* of a finished candidate, not a step in producing one. The theorem below applies to exactly this class of rules and to no wider one.

Every class of the classification is a rule. Class A emits  $C = \{a\}$  on qualified transactions and  $C = \emptyset$  otherwise. Class E emits the roots of (10) for each admissible branch of  $\sigma$ , discarding those outside  $[1, n - 1]$ . Class B emits  $\{am\}$  for each integer root  $m$  of (5). Uniform guessing is the rule that emits  $t$  i.i.d. uniform draws.

**Theorem 7** (Zero yield). Under Assumptions A and B — equivalently, for every abscissa-oblivious rule  $R$ , for which Assumption B is by Definition 6 no restriction —

$$\Pr[\zeta \in C] = \frac{\mathbb{E}[|C \cap (\mathbb{Z}_n^* \setminus \{s\})|]}{n - 2} \leq \frac{\mathbb{E}[|C|]}{n - 2}. \quad (3)$$

In particular the success probability per candidate emitted is the constant  $1/(n - 2)$ , independent of  $R$ .

*Proof.* By Lemma 4(ii) the triple  $(\zeta, s, s_{zr})$  is uniform on the product set  $\mathcal{S}$ . Hence, conditionally on the public data — a function of  $(s, s_{zr}, r)$ , all independent of  $\zeta$  by Lemma 4(iii) — the hidden component  $\zeta$  is uniform on  $\mathbb{Z}_n^* \setminus \{s\}$ , a set of exactly  $n - 2$  elements. The set  $C$  is measurable with respect to that same public data together with auxiliary randomness independent of the signer, hence conditionally fixed. For a uniform variable on a set of size  $n - 2$  and a fixed subset  $C$ ,  $\Pr[\zeta \in C \mid \text{public}] = |C \cap (\mathbb{Z}_n^* \setminus \{s\})|/(n - 2)$ . Take expectations.  $\square$

**Corollary 8** (No-go for abscissa-oblivious rules). No abscissa-oblivious classification of ECDSA signatures can achieve a per-candidate success rate different from  $1/(n - 2)$ . Consequently:

- (i) no rule is better than uniform guessing on  $\mathbb{Z}_n^* \setminus \{s\}$ ; equality in (3) holds for that rule and for every rule emitting distinct candidates;
- (ii) a rule can be worse only by emitting duplicates or the excluded value  $s$ , both implementation defects rather than properties of the class;
- (iii) against a guesser drawing uniformly from all of  $\mathbb{Z}_n^*$ , whose yield is  $1/(n - 1)$ , the best possible rule wins by the factor  $(n - 1)/(n - 2) = 1 + 1/(n - 2)$ , equal to  $1 + 8.64 \times 10^{-78}$  on secp256k1. That advantage is purchased entirely by the observation  $\zeta \neq s$ , which requires no algebra.

**Corollary 9** (The boundary is ECDLP, not information). *A rule whose per-candidate yield exceeds  $1/(n-2)$  must fail to be abscissa-oblivious: its output must depend on the true  $r = [kG]_x$  in a way that an independent uniform  $r$  would not reproduce. Any such dependence is a use of the group law of  $E$ . Since  $r$  determines  $R = \pm kG$  and the step from  $R$  to  $k$  is by definition a discrete logarithm, such a rule solves an instance of ECDLP.*

*Proof.* Immediate from Definition 6 and Theorem 7. If the rule’s output law is invariant under replacing  $r$  by an independent uniform draw, the theorem applies and the yield is  $1/(n-2)$ . Contrapositively, a yield above  $1/(n-2)$  forces non-invariance, that is, genuine dependence on the abscissa map  $k \mapsto [kG]_x$ .  $\square$

**Remark 10** (What is *not* being claimed). ECDSA offers no information-theoretic security, and nothing above suggests otherwise. A single signature determines the private key up to a handful of candidates:  $r$  fixes the abscissa of  $R = kG$ , the curve equation gives  $R$  up to sign, and  $k$  follows, whence  $x = (sk - z)r^{-1}$ . On secp256k1 the additional possibility  $[R]_x = r + n$  arises with probability  $(p - n)/n \approx 2^{-128}$ , so the candidate count is essentially two. The residual entropy of  $x$  given  $(z, r, s)$  is therefore one or two bits, not 256 — which is precisely why public-key recovery from a signature is a routine operation. The hardness of ECDSA is computational and resides entirely in ECDLP.

Assumption B deletes that structure. Doing so is what makes Theorem 7 provable, and it is equally what confines the theorem to abscissa-oblivious rules. The theorem is worth stating because classes A, B, E — and every classification of the same shape — lie inside that confinement by construction. It would be a serious error to read it as a claim that ECDSA signatures leak nothing.

**Remark 11.** The correct negative statement is an equality, not an inequality. “Worse than guessing” would suggest an inefficiency that a cleverer rule might remove; “exactly equal to guessing, for every rule” is a conservation law, and there is nothing left to optimise. Sections 5 through 8 may therefore be read as a computation of the single scalar  $\mathbb{E}|C|$  for three particular rules, undertaken because the classes were already in circulation and their frequencies had been misreported.

## 5 The qualification event

**Definition 12** (Qualification). A transaction *qualifies*, written  $\mathcal{Q}$ , if  $s_{zr} > s$  and the residue  $a = s \bmod d$  satisfies  $a > 0$ ,  $a \neq s$  and  $a \neq s_{zr}$ .

**Lemma 13** (Geometric form).  $\mathcal{Q}$  holds if and only if  $s < s_{zr} \leq 2s$  and  $d \nmid s$ .

*Proof.* Assume  $s_{zr} > s$ , so  $d \geq 1$ . If  $d > s$  then  $s \bmod d = s$ , violating  $a \neq s$ ; conversely if  $d \leq s$  then  $a < d \leq s$ , so  $a \neq s$ . Thus  $a \neq s \iff d \leq s \iff s_{zr} \leq 2s$ . Next,  $a = 0 \iff d \mid s$ . Finally  $a < d < s_{zr}$  always, so  $a \neq s_{zr}$  is automatic.  $\square$

**Theorem 14** (Qualification rate).  $\Pr[\mathcal{Q}] = \frac{1}{4} - \Theta(n^{-1} \log n) \rightarrow \frac{1}{4}$ , independently of the curve, the field and the key.

*Proof.* By Lemma 4(iii) the pair  $(s, s_{zr})$  is, up to an  $O(1/n)$  perturbation of the marginal of  $s$ , uniform on  $\{1, \dots, n-1\}^2$ . Write  $M = n-1$ , which is even for every odd prime  $n$ , and count lattice points with  $s < s_{zr} \leq \min(2s, M)$ :

$$\sum_{s=1}^M (\min(2s, M) - s) = \underbrace{\sum_{s=1}^{M/2} s}_{= M^2/8 + M/4} + \underbrace{\sum_{s=M/2+1}^M (M - s)}_{= M^2/8 - M/4} = \frac{M^2}{4},$$

an exact identity. The excluded set  $d \mid s$  contributes  $\sum_{s \leq M} \tau(s) = M \log M + (2\gamma - 1)M + O(\sqrt{M})$  points by Dirichlet’s divisor theorem [6, Thm. 320]. Dividing by  $M^2$  gives  $\Pr[\mathcal{Q}] = \frac{1}{4} - (\log M + 2\gamma - 1)/M + O(M^{-1/2})$ , and the perturbation of the marginal of  $s$  adds a further  $O(1/M)$ .  $\square$

**Remark 15.** The often-quoted “25% anomaly rate” of this project is thus the area of the region  $s < s_{zr} \leq 2s$  in the unit square. It is scale-invariant because that area is  $1/4$  for every  $n$ . It measures the geometry of comparing two independent residues and carries no information about key recoverability; its constancy across curves is a reason to expect no exploitability, not a reason to hope for it.

## 6 Class A: linear divisibility with unit multiplier

**Definition 16** (Class A). A qualified transaction is in class A if the ratio  $m_1 = \zeta/a$ , computed in  $\mathbb{Q}$ , equals 1; equivalently  $\zeta = a$  as integers. (The strict taxonomy additionally demands  $m_2 = (s + s_{zr})/\zeta \notin \mathbb{Z}$ , separating A from B; by Proposition 22 that refinement removes an  $O(n^{-2} \log n)$  fraction and is invisible at every scale considered here.)

**Theorem 17** (Recovery). *On a class-A transaction,  $k = az^{-1} \bmod n$  and  $x = (s - a)(rk)^{-1} = z(s - a)(ra)^{-1} \bmod n$ . One candidate is produced, correct with certainty, verified by one scalar multiplication  $xG \stackrel{?}{=} Q$ .*

*Proof.* Substitute  $\hat{s} = a = \zeta$  into Proposition 3(i). All inverses exist because  $n$  is prime and  $z, r, a \neq 0$ .  $\square$

**Theorem 18** (Exact frequency). *For every prime  $n \geq 5$ ,*

$$\Pr[A] = \frac{\Pr[Q]}{n-2} = \frac{1}{4(n-2)} \left(1 - \Theta(n^{-1} \log n)\right) \sim \frac{1}{4n}, \quad (4)$$

*an identity in the model of Lemma 4, not an asymptotic estimate. Equivalently, the class-A rule emits  $\mathbb{E}[C] = \Pr[Q]$  candidates per signature.*

*Proof.* Condition on  $(s, s_{zr})$ . If it does not qualify,  $C = \emptyset$ . If it qualifies then  $d$  and  $a$  are determined, and by Lemma 13  $1 \leq a < d \leq s$ , so  $a \in \mathbb{Z}_n^* \setminus \{s\}$  and  $|C| = 1$ . Apply Theorem 7.  $\square$

*Erratum 1* (A wrong turn worth recording). It is tempting to compute  $\Pr[A]$  by counting, for each  $(s, \zeta)$ , the divisors  $d \mid (s - \zeta)$  with  $d > \zeta$ , and approximating the resulting sum by an integral. That route yields  $\Pr[A] \approx 0.3466/n$ , which is 39% too large and is rejected by the tiny-curve data at 16.8 $\sigma$ . The continuum approximation silently discards the fractional parts  $\{(n - \zeta)/d\}$ , whose systematic mean of  $\frac{1}{2}$  accumulates to a term of order  $d$  under the  $\zeta$ -summation. Carrying the floors exactly gives  $\sum_{\zeta=0}^{d-1} \lfloor (M - \zeta)/d \rfloor = M - d + 1$  and restores  $M^2/4$ , in agreement with Theorem 18. Lattice counts in this subject are dominated by their floor terms; integral approximations to them should be distrusted by default.

## 7 Class B: dual integral divisibility

**Definition 19** (Class B). A qualified transaction is in class B if  $m_1 = \zeta/a$  and  $m_2 = (s + s_{zr})/\zeta$  are both integers and  $m_1 = m_2 =: m$ . Both quotients are taken in  $\mathbb{Q}$ : they assert exact divisibility over  $\mathbb{Z}$ , not congruences.

**Theorem 20** (The multiplier satisfies an integer quadratic). *On a class-B transaction, with  $\sigma \in \{s, s + n\}$  the true integer sum (2),*

$$am^2 - \sigma m + (s + s_{zr}) = 0 \quad \text{in } \mathbb{Z}. \quad (5)$$

*Hence  $\Delta = \sigma^2 - 4a(s + s_{zr})$  is a perfect square in  $\mathbb{Z}$  and  $m = (\sigma \pm \sqrt{\Delta})/(2a) \in \mathbb{Z}$ , giving  $\zeta = am$  and  $x = z(s - \zeta)(r\zeta)^{-1} \bmod n$ . At most four candidates arise, two roots for each branch of  $\sigma$ .*

*Proof.* By definition  $\zeta = am$  and  $s + s_{zr} = m\zeta$  over  $\mathbb{Z}$ . By (2),  $\zeta = \sigma - \zeta = \sigma - am$ , so  $s + s_{zr} = m(\sigma - am) = \sigma m - am^2$ , which is (5). All coefficients are integers, so the quadratic formula over  $\mathbb{Z}$  forces  $\Delta$  to be a perfect square.  $\square$

**Lemma 21** (Wrap-around in class B). *If a class-B transaction has  $\sigma = s + n$ , then  $m \leq 2$ ; moreover  $m = 1$  implies  $s_{zr} + a = n$ , and  $m = 2$  implies that  $d$  is even and  $4a + d = 2n$ .*

*Proof.* Qualification gives  $d \leq s$ , so  $s + s_{zr} = 2s + d \leq 3s$ . Also  $\zeta = s + n - \zeta \geq s + 1$  because  $\zeta \leq n - 1$ . From  $m\zeta = s + s_{zr}$  we get  $m \leq 3s/(s + 1) < 3$ . For  $m = 1$ :  $\zeta = 2s + d$  and  $\zeta = s + n - a$  give  $n = s + d + a = s_{zr} + a$ . For  $m = 2$ :  $2\zeta = 2s + d$  forces  $d$  even with  $\zeta = s + d/2$ , while  $\zeta = s + n - 2a$ ; equating gives  $2n = 4a + d$ .  $\square$

Each is one exact equation between quantities of size  $\Theta(n)$ , hence a codimension-one condition on top of an already  $O(n^{-2})$ -rare event. No wrap-around instance occurred in the exhaustive enumeration of Section 9. Implementations should nonetheless test both branches; the cost is one extra scalar multiplication.

**Proposition 22** (Frequency of class B). *Under a divisor-equidistribution heuristic,*

$$\Pr[B] = \Theta\left(\frac{\log n}{n^2}\right), \quad \text{empirically} \quad \Pr[B] \approx \frac{c_B \log n}{n^2}, \quad c_B \approx 0.19. \quad (6)$$

*Argument.* Fix  $(s, s_{zr})$  qualified. Each divisor  $m$  of  $s + s_{zr}$  determines a unique candidate pair  $\zeta = am$ ,  $\xi = (s + s_{zr})/m$ ; the pair survives only if additionally  $\zeta + \xi \in \{s, s + n\}$ , a coincidence of probability  $\Theta(1/n)$  under equidistribution. The mean of  $\tau$  on  $[1, 2n]$  is  $\log n + O(1)$ , so the conditional rate is  $\Theta(n^{-1} \log n) \cdot n^{-1}$ ; multiplying by  $\Pr[\mathcal{Q}] = \frac{1}{4}$  gives  $\approx \frac{1}{4} n^{-2} \log n$ . Exhaustive enumeration for  $101 \leq n \leq 1009$  yields  $\Pr[B] n^2 / \log n \in [0.189, 0.196]$ , so  $c_B \approx 0.19$ , within 25% of the heuristic constant  $\frac{1}{4}$ .  $\square$

*Remark 23* (Weakest link, stated plainly). The measured ratio drifts downward by about 3% across the accessible range, so the true law may carry a sub-logarithmic correction that the data cannot resolve; the accessible range spans one decade of  $n$  while the extrapolation target spans 74. This is the least secure claim in the paper. It is also inconsequential: an error of two orders of magnitude in  $c_B$  moves  $\Pr[B]$  on secp256k1 from  $10^{-153}$  to somewhere between  $10^{-155}$  and  $10^{-151}$ , and Theorem 7 is unaffected either way. Classes A and E carry no such caveat: their formulas are exact for every prime  $n$ .

## 8 Class E: the “Goldbach-type” decomposition

Class E is the most instructive of the three: its published description is ornate and its actual content is one line.

**Definition 24** (Classes E1 and E2). With  $\sigma = \zeta + \xi$  the integer sum (2), a transaction is in class E if  $\sigma$  is even and

$$|\zeta - \xi| = \frac{\sigma}{2} + 1. \quad (7)$$

Instances with  $\sigma = s$  are labelled E1 and those with  $\sigma = s + n$  are labelled E2. The published formulation writes  $\lfloor \sigma/2 \rfloor + 1$ ; since  $\sigma$  is required to be even, the floor is vacuous.

**Theorem 25** (Class E is an affine line). *For integers  $\zeta, \xi \geq 1$ , condition (7) together with “ $\sigma$  even” holds if and only if*

$$\zeta = 3\xi + 2 \quad \text{or} \quad \xi = 3\zeta + 2 \quad \text{in } \mathbb{Z}. \quad (8)$$

*Moreover every class-E instance satisfies  $\sigma \equiv 2 \pmod{4}$ , and the two branches are disjoint.*

*Proof.* ( $\Rightarrow$ ) Equality (7) rules out  $\zeta = \xi$ , its right-hand side being positive. Take  $\zeta > \xi$  without loss of generality; then  $\zeta - \xi = (\zeta + \xi)/2 + 1$ , i.e.  $2\zeta - 2\xi = \zeta + \xi + 2$ , i.e.  $\zeta = 3\xi + 2$ . ( $\Leftarrow$ ) If  $\zeta = 3\xi + 2$  then  $\sigma = 4\xi + 2$  is even and  $\zeta - \xi = 2\xi + 2 = \sigma/2 + 1$ , so (7) holds; and  $\sigma = 4\xi + 2$  gives  $\sigma \equiv 2 \pmod{4}$ . If both branches held,  $\zeta = 3(3\zeta + 2) + 2 = 9\zeta + 8$  forces  $\zeta = -1$ , impossible for a positive integer.  $\square$

*Remark 26* (The Goldbach framing is empty). Theorem 25 dissolves the name. The published description — “a symmetric pair around  $\sigma/2$  with fixed difference” — is doubly vacuous: *every* pair summing to  $\sigma$  is symmetric about  $\sigma/2$  by construction, and the only content of (7) is that the hidden pair lies on one particular line of slope 3. There is no additive number theory here and no connection to the Goldbach problem beyond the superficial fact that a number is written as a sum of two others.

**Theorem 27** (Exact cardinality and frequency). *For every prime  $n \geq 5$  the number of class-E hidden pairs in  $\{1, \dots, n-1\}^2$  is*

$$\#E = 2 \left\lfloor \frac{n-3}{3} \right\rfloor, \quad \text{so} \quad \Pr[E] = \frac{2 \lfloor (n-3)/3 \rfloor}{(n-1)^2} = \frac{2}{3n} + O\left(\frac{1}{n^2}\right). \quad (9)$$

*Proof.* By Theorem 25 the class is a disjoint union of two branches. On  $\zeta = 3\tilde{\zeta} + 2$  the constraints are  $\tilde{\zeta} \geq 1$  and  $3\tilde{\zeta} + 2 \leq n - 1$ , i.e.  $1 \leq \tilde{\zeta} \leq (n - 3)/3$ , giving  $\lfloor (n - 3)/3 \rfloor$  solutions; the other branch gives the same count by symmetry. Class E depends only on  $(\zeta, \tilde{\zeta})$ , uniform on  $(\mathbb{Z}_n^*)^2$  by Lemma 4(i), a set of  $(n - 1)^2$  elements. No class-E pair is removed by  $s \neq 0$ , since  $\sigma = 4\tilde{\zeta} + 2 = n$  would force  $n$  even.  $\square$

**Theorem 28** (Recovery). *Given the effective sum  $\sigma \in \{s, s + n\}$  with  $\sigma \equiv 2 \pmod{4}$ , put*

$$\alpha = \frac{3\sigma + 2}{4}, \quad \beta = \sigma - \alpha = \frac{\sigma - 2}{4}. \quad (10)$$

*Then  $\{\zeta, \tilde{\zeta}\} = \{\alpha, \beta\}$ , and  $x$  is one of  $x_c = z(s - c)(rc)^{-1} \pmod{n}$  for  $c \in \{\alpha, \beta\}$ , disambiguated by one scalar multiplication.*

*Proof.* Solve  $\zeta + \tilde{\zeta} = \sigma$ ,  $\zeta - \tilde{\zeta} = \sigma/2 + 1$ : adding gives  $2\zeta = 3\sigma/2 + 1$ , i.e.  $\zeta = (3\sigma + 2)/4$ , an integer precisely because  $\sigma \equiv 2 \pmod{4}$  (Theorem 25). Apply Proposition 3(i) to each candidate.  $\square$

**Theorem 29** (Offset invariance). *For any fixed  $c \in \mathbb{Z}$  define  $E_c$  by “ $\sigma$  even and  $|\zeta - \tilde{\zeta}| = \sigma/2 + c$ ”. Then  $E_c = \{\zeta = 3\tilde{\zeta} + 2c\} \sqcup \{\tilde{\zeta} = 3\zeta + 2c\}$ , and for  $|c| = o(n)$ ,*

$$\Pr[E_c] = \frac{2}{3n} \left(1 + O(|c|/n)\right),$$

*independently of  $c$ . The Goldbach value  $c = 1$  is thus one member of a family of  $\Theta(n)$  equally frequent classes.*

*Proof.* Repeat Theorem 25 with right-hand side  $\sigma/2 + c$ : for  $\zeta > \tilde{\zeta}$ ,  $2\zeta - 2\tilde{\zeta} = \zeta + \tilde{\zeta} + 2c$  gives  $\zeta = 3\tilde{\zeta} + 2c$ ; conversely  $\sigma = 4\tilde{\zeta} + 2c$  is even and  $\zeta - \tilde{\zeta} = 2\tilde{\zeta} + 2c = \sigma/2 + c$ . Counting on one branch,  $\#\{\tilde{\zeta} : 1 \leq \tilde{\zeta} \leq n - 1, 1 \leq 3\tilde{\zeta} + 2c \leq n - 1\} = n/3 + O(|c|)$ ; the other is symmetric.  $\square$

A control experiment in the project repository reports a flat frequency of  $\approx 0.67/n$  for offsets  $c \in \{\pm 1, 3, 5, 7, 9, 101\}$ . Theorem 29 accounts for it exactly: the constant  $0.67 \approx 2/3$  is  $\Pr[E_c] \cdot n$ , and its independence of  $c$  is a two-line consequence of the affine structure. An experiment run to test a hypothesis turns out to be a corollary of that hypothesis’s own algebra.

## 8.1 Class versus detector

The rule of Definition 5 associated with class E must try both branches of  $\sigma$ , since it cannot see which is real. Counting admissible candidates:  $\sigma = s$  contributes two whenever  $s \equiv 2 \pmod{4}$ ;  $\sigma = s + n$  contributes two when additionally  $s \leq n/3$  and one otherwise, because  $\alpha$  then exceeds  $n - 1$  while  $\beta$  does not. Hence

$$\mathbb{E}|C_E| = \frac{1}{4} \cdot 2 + \frac{1}{4} \left( \frac{1}{3} \cdot 2 + \frac{2}{3} \cdot 1 \right) = \frac{5}{6}, \quad \Pr[\text{rule E succeeds}] = \frac{5}{6(n - 2)}, \quad (11)$$

which exceeds  $\Pr[E] = 2/(3n)$  of Theorem 27. There is no contradiction: the *class* is defined using the secret  $\sigma$ , whereas the *rule* also scores accidental hits from the wrong branch, and the gap  $5/6$  against  $2/3$  is exactly their weight. This distinction — between a class defined with oracle access to the secret and a rule that must operate without it — is where informal treatments of the subject usually go wrong.

## 9 Numerical validation

### 9.1 Exhaustive enumeration

For small  $n$  the entire state space  $S$  of Lemma 4, of size  $(n - 1)^3$ , can be enumerated without sampling, giving exact probabilities. Table 1 reports all states for seven primes,  $1.66 \times 10^9$  in total.

### 9.2 Independently generated test-curve data

The datasets accompanying the earlier framework were generated on genuine elliptic curves, so Assumption B played no role in producing them. Table 2 compares those counts with the formulas proved above. All residuals lie within Poisson fluctuation; the largest,  $+2.05\sigma$  on 12 observed events, has a one-sided  $p$ -value of 0.05. Measured qualification rates were 0.2494 and 0.2503 on the two million-transaction datasets, against  $1/4$  predicted.

| $n$  | states | $\Pr[\mathcal{Q}]$ | $\Pr[\mathcal{A}]$ | $\Pr[\mathcal{Q}]/(n-2)$ | #E  | $2\lfloor \frac{n-3}{3} \rfloor$ | $\Pr[\mathcal{B}]n^2/\ln n$ |
|------|--------|--------------------|--------------------|--------------------------|-----|----------------------------------|-----------------------------|
| 101  | 1.00e6 | 0.2096820          | 2.118000e-3        | 2.118000e-3              | 64  | 64                               | 0.192                       |
| 151  | 3.38e6 | 0.2205200          | 1.480000e-3        | 1.480000e-3              | 98  | 98                               | 0.195                       |
| 251  | 1.56e7 | 0.2303389          | 9.250560e-4        | 9.250560e-4              | 164 | 164                              | 0.196                       |
| 401  | 6.40e7 | 0.2364823          | 5.926875e-4        | 5.926875e-4              | 264 | 264                              | 0.193                       |
| 503  | 1.26e8 | 0.2388253          | 4.766967e-4        | 4.766967e-4              | 332 | 332                              | 0.194                       |
| 751  | 4.22e8 | 0.2419628          | 3.230483e-4        | 3.230483e-4              | 498 | 498                              | 0.191                       |
| 1009 | 1.03e9 | 0.2437219          | 2.420277e-4        | 2.420277e-4              | 670 | 670                              | 0.189                       |

**Table 1:** Exhaustive enumeration of the full state space. Columns 4–5 confirm Theorem 18 as an identity to every displayed digit; columns 6–7 confirm (9) exactly; the last column calibrates  $c_B \approx 0.19$  in (6).  $\Pr[\mathcal{Q}]$  approaches  $1/4$  from below at the rate  $\Theta(\log n/n)$  of Theorem 14.

| curve | $n$           | $N_{\text{tx}}$ | A pred  | A obs | $z_A$ | B pred | B obs | E pred  | E obs | $z_E$ |
|-------|---------------|-----------------|---------|-------|-------|--------|-------|---------|-------|-------|
| tiny  | 9 967         | 9.93e7          | 2 491.8 | 2 465 | −0.54 | 1.75   | 2     | 6 642.2 | 6 746 | +1.27 |
| small | 99 667        | 1.00e6          | 2.51    | 4     | +0.94 | 2e-4   | 0     | 6.69    | 12    | +2.05 |
| large | 1 241 630 743 | 1.00e6          | 2e-4    | 0     | −0.01 | 3e-12  | 0     | 5e-4    | 0     | −0.02 |

**Table 2:** Predicted versus observed counts on the project’s test curves.

### 9.3 Monte-Carlo cross-check and the yield constant

Simulating the model of Lemma 4 with  $2.4 \times 10^7$  draws per curve gave, at  $n = 9967$ : qualification rate 0.24909;  $\#A = 600$  against 599.8 predicted;  $\#E = 1614$  against 1605.3. At  $n = 99667$ : qualification 0.24978,  $\#A = 71$  (predicted 60.1),  $\#E = 177$  (predicted 160.5). In all  $4.8 \times 10^7$  draws the number of transactions satisfying Definition 24 was *identical* to the number satisfying the affine criterion (8) —  $1614 = 1614$  and  $177 = 177$  — confirming Theorem 25 without exception.

Table 3 tests Theorem 7 directly by measuring, for each rule, both the expected candidate count and the empirical yield per candidate.

## 10 Evaluation on secp256k1

The group order is the prime

$$n = \text{FFFFFFFF FFFFFFFF FFFFFFFF FFFFFFFF BAAEDCE6 AF48A03B BFD25E8C D0364141}_{16} \\ \approx 1.157920892 \times 10^{77},$$

with  $\log_2 n = 256.000000$ ,  $\ln n = 177.4457$ ,  $n \equiv 1 \pmod{3}$  and  $n \equiv 1 \pmod{4}$ . Because  $n \equiv 1 \pmod{3}$  the floor in (9) evaluates exactly to  $(n-4)/3$ , so no approximation enters the class-A or class-E figures.

Taking  $N_{\text{tx}} = 10^{10}$  as a generous upper bound on all secp256k1 signatures ever produced — Bitcoin has processed on the order of  $10^9$  transactions carrying a few  $\times 10^9$  input signatures, Ethereum a few  $\times 10^9$  transactions — the expected number of instances of all three classes combined is  $7.9 \times 10^{-68}$ ; at  $N_{\text{tx}} = 10^{12}$  it is  $7.9 \times 10^{-66}$ , and at an absurd  $10^{20}$  it is  $7.9 \times 10^{-58}$ . Since these expectations are far below 1, Markov’s inequality makes them upper bounds on the probability that even one such signature exists.

The sample size for even odds of a single occurrence is  $N_{1/2} = \ln 2 / \Pr$ :  $1.20 \times 10^{77}$  for E,  $3.21 \times 10^{77}$  for A,  $8.76 \times 10^{76}$  for the union and  $2.76 \times 10^{152}$  for B. Every entry exceeds the group order.

**Abundance without density.** The classes are not empty on secp256k1. The transaction space  $\mathcal{T} = \{1, \dots, n-1\}^4$  has  $|\mathcal{T}| \approx 1.80 \times 10^{308}$  elements, of which  $\#A \approx 3.88 \times 10^{230}$ ,  $\#E \approx 1.04 \times 10^{231}$  and  $\#B \approx 4.5 \times 10^{155}$ . These are enormous absolute counts and a fraction  $\approx 10^{-77}$  of the space. An analyst who reasons “there are  $10^{231}$  vulnerable signatures out there” has said something true and entirely without consequence, in the same way that there are  $10^{231}$  private keys beginning with any particular 256-bit prefix. Absolute counts in this subject should never be quoted without the density beside them.

*Remark 30* (Physical bound, for scale only). Landauer’s principle [9] charges at least  $k_B T \ln 2 = 2.61 \times 10^{-23}$  J per irreversible bit operation at the cosmic microwave background temperature. Even at one

| rule    | $\mathbb{E} C $ predicted          | $\mathbb{E} C $ measured | yield/candidate measured | $1/(n-2)$              |
|---------|------------------------------------|--------------------------|--------------------------|------------------------|
| class A | $\Pr[\mathcal{Q}] \rightarrow 1/4$ | 0.2489                   | $1.031 \times 10^{-4}$   | $1.004 \times 10^{-4}$ |
| class E | 5/6                                | 0.8325                   | $9.809 \times 10^{-5}$   | $1.004 \times 10^{-4}$ |
| class B | $\approx 0.19 \ln n/n$             | $0.184 \ln n/n$          | —                        | $1.004 \times 10^{-4}$ |

**Table 3:** Direct test of Theorem 7 at  $n = 9\,967$  over  $3 \times 10^6$  simulated signatures. The yield per candidate is the same constant for every rule, as the theorem requires; class B emitted too few candidates at this sample size for a yield estimate.

| event             | closed form                                 | value                     | power of 2           | status               |
|-------------------|---------------------------------------------|---------------------------|----------------------|----------------------|
| $\mathcal{Q}$     | $\frac{1}{4} + O(n^{-1} \log n)$            | 0.250000...               | $2^{-2.000}$         | theorem              |
| A                 | $\Pr[\mathcal{Q}]/(n-2)$                    | 2.159042e-78              | $2^{-258.00}$        | exact identity       |
| B                 | $\approx 0.19 \ln n/n^2$                    | $\approx 2.5\text{e-}153$ | $\approx 2^{-506.9}$ | calibrated heuristic |
| E                 | $2 \lfloor \frac{n-3}{3} \rfloor / (n-1)^2$ | 5.757446e-78              | $2^{-256.58}$        | exact identity       |
| $A \cup B \cup E$ | union bound                                 | 7.916488e-78              | $2^{-256.24}$        | —                    |
| one blind guess   | $1/(n-1)$                                   | 8.636169e-78              | $2^{-256.00}$        | exact                |

**Table 4:** Per-signature probabilities on secp256k1. These are per-signature figures; the like-for-like comparison at equal numbers of tested candidates is Corollary 8.

bit erasure per trial, enumerating  $1/\Pr[A] = 4.63 \times 10^{77}$  trials costs  $1.21 \times 10^{55}$  J, the mass–energy of  $6.8 \times 10^7$  solar masses. The figure is quoted once, for scale, and no argument in this paper rests on it: Corollary 8 already settles the matter at the level of algorithms, where it belongs.

## 11 Errata to previously circulated material

This project circulated earlier write-ups of classes A and B under a “vulnerability” framing, together with a framework document stating frequency formulas for all five classes. That framing is withdrawn, and three specific numerical or structural claims are corrected here.

*Erratum 2* (The class-E rate is  $2/(3n)$ , not  $1/(n-1)$ ). The framework document states  $\Pr[E] = 1/(n-1)$ , reasoning that the condition “fixes  $s_{zk}$  to at most two values” and that “exactly one of  $s, s+n$  is even”. The second clause is correct; the first over-counts. By Theorem 25 an even  $\sigma$  admits a solution only when  $\sigma \equiv 2 \pmod{4}$ , which removes half the even sums, and the surviving locus has length  $\approx n/3$  per branch rather than  $\approx n/2$ . The correct rate is  $2/(3n)$ , that is,  $\frac{2}{3}$  of the published value.

The distinction is not academic. With  $n = 9\,967$  and  $N_{\text{tx}} = 99\,323\,922$  — the project’s own tiny-curve dataset, published alongside the formula — the model  $1/(n-1)$  predicts 9966.5 events against 6746 observed, a Poisson deviation of  $-32.3\sigma$ ; Theorem 27 predicts 6642.2, a deviation of  $+1.27\sigma$ . The published formula was excluded by the data accompanying it.

*Erratum 3* (No modular square root is required in class B). The published statement of the class-B recovery reduces the quadratic modulo  $n$ , computes  $\sqrt{\Delta} \bmod n$  via the Euler criterion, and inverts  $2a$  modulo  $n$ . That procedure is not incorrect, but it is strictly weaker and needlessly expensive: it discards the fact that (5) holds over  $\mathbb{Z}$ , where “ $\Delta$  is a perfect square” is far more restrictive than “ $\Delta$  is a quadratic residue mod  $n$ ” (density  $\Theta(\Delta^{-1/2})$  against  $1/2$ ). It also fixes  $\sigma = s$ , silently assuming no wrap-around; see Lemma 21. In exhaustive enumeration for  $n \leq 1009$  — 3823 class-B instances — the integer discriminant was a perfect square in every case.

*Erratum 4* (Correction to an earlier draft of this paper). An earlier draft compared an “anomaly hunter” processing  $T$  signatures against a guesser making  $3T$  guesses, concluding that the hunter loses by a factor of 3.3. The arithmetic is right but the accounting is arbitrary: it charges the hunter for three candidates per signature, whereas the classes emit  $\mathbb{E}|C| \approx \frac{1}{4} + \frac{5}{6} + O(n^{-1} \log n) \approx 1.083$ . Charging both adversaries the same number of scalar multiplications gives an *exact tie*, up to the factor  $1 + 1/(n-2)$  of Corollary 8. The corrected statement is stronger, for the reason given in Remark 11.

## 12 Discussion

### 12.1 Falsifiability

The results are conditional on Assumptions A and B. The useful question is what observation would overturn them. A class whose observed frequency exceeds its predicted  $c/n$  baseline by a factor  $\lambda > 1$  that persists as  $n$  grows would demonstrate a genuine non-uniformity in the law of  $\zeta$  — the only finding in this area worth pursuing, since it would contradict Lemma 4 and hence Theorem 7.

Such a test is cheap on test curves. For a Poisson count with baseline rate  $p_0$ , distinguishing  $p_0$  from  $\lambda p_0$  at significance  $\alpha$  and power  $1 - \beta$  requires

$$N \gtrsim \frac{(z_{\alpha/2} + z_{\beta})^2}{p_0(\sqrt{\lambda} - 1)^2}. \quad (12)$$

At  $\alpha = 0.05$  and 80% power, detecting  $\lambda = 2$  in class E on the tiny curve needs  $6.8 \times 10^5$  transactions;  $\lambda = 1.10$  needs  $4.9 \times 10^7$ ;  $\lambda = 1.01$  needs  $4.7 \times 10^9$ . The existing  $9.9 \times 10^7$ -transaction dataset already resolves a 10% enrichment.

Applying (12) to the data in hand: the tiny-curve class-E count (6746 against 6642.2 predicted) constrains  $\lambda$  to  $1.016 \pm 0.012$ , excluding any enrichment above 4% at 95% confidence; the class-A count (2465 against 2491.8) constrains  $\lambda$  to  $0.989 \pm 0.020$ . Both are consistent with  $\lambda = 1$  to within a couple of percent.

*Remark 31* (The one experiment worth running). Searching for further coincidence classes is not it: the space of arithmetic coincidences is infinite and, by Theorem 7, each new class will have frequency  $\mathbb{E}|C|/(n-2)$  with  $\mathbb{E}|C| = \Theta(1)$  at best. The informative experiment is a direct test of the *distribution*: take  $10^8$  to  $10^9$  transactions on a curve with  $n \approx 10^4$ – $10^5$ , form the empirical law of  $\zeta = zk^{-1}$  conditioned on every public statistic available, and test uniformity by a chi-squared statistic over  $\sqrt{N}$  bins or by a discrete Fourier test in the manner of Bleichenbacher. A null result closes the question; a positive one would announce itself as a bias in a histogram, not as an elegant identity.

### 12.2 Limitations

1. **Assumption B is a heuristic.** Treating  $r$  as an independent uniform unit is not provable with current techniques. Its role is confined to obtaining closed forms, and the predictions were checked against data generated without it (Table 2).
2. **The class-B constant is empirical.**  $c_B \approx 0.19$  is calibrated over  $101 \leq n \leq 1009$  and drifts by  $\approx 3\%$  across that range. Only the order  $\Theta(n^{-2} \log n)$  is argued from first principles, and that rests on divisor equidistribution.
3. **Ideal signer only.** Nothing here bears on implementations with faulty randomness, deterministic-nonce bugs, fault injection or side channels; those are the real attack surface and are governed by the works cited in Section 1.4.
4. **Union bounds.** The three classes share the same hidden pair and are not independent, so the union figures in Table 4 are upper bounds — the conservative direction.
5. **Classes C and D are out of scope.** Class C admits no closed-form recovery. Class D is not a coincidence class but the generic case, occurring for  $\approx 25\%$  of transactions by Theorem 14; its candidate window has width  $a = \Theta(n)$  on average, so  $\mathbb{E}|C_D| = \Theta(n)$  and Theorem 7 gives success probability  $\Theta(1)$  at cost  $\Theta(n)$  — that is, exhaustive search, roughly  $2^{128}$  times worse than Pollard rho. The level index  $N = \lfloor \zeta/a \rfloor$  is moreover a function of the secret  $\zeta$ , so the window cannot be located without already knowing what it is meant to reveal.

## 13 Conclusion

Under an ideal signer the hidden pair is uniform on  $(\mathbb{Z}_n^*)^2$  and, jointly with the public  $s_{Zr}$ , uniform on a product set. Conditioning on all public data therefore leaves  $\zeta$  uniform over  $n - 2$  values, and every subsequent result is a consequence of that one sentence: the yield per tested candidate is the constant  $1/(n - 2)$  for every rule computable from public data, so a classification can change only how

many candidates are emitted, never their quality. The best conceivable rule of this type beats uniform random key guessing by the factor  $1 + 8.6 \times 10^{-78}$ , and exhaustive key search is itself  $\approx 2^{128}$  times more expensive than the generic attack secp256k1 was designed to resist.

Within that frame the three classes are exercises. Class A is the graph of a function and occurs with probability  $\Pr[Q]/(n-2)$ ; class E is the affine line  $\zeta = 3\tilde{\zeta} + 2$  and occurs with probability  $2\lfloor(n-3)/3\rfloor/(n-1)^2$ ; class B is a divisor coincidence of order  $n^{-2} \log n$  whose multiplier satisfies an integer quadratic. The qualification rate is the area of a lattice region and equals  $1/4$  on every curve. The distinguished offset of class E is one of  $\Theta(n)$  equivalent choices.

The productive question is not whether more classes exist — infinitely many do, all of the same quality — but whether the distribution of  $s_{zk}$  is uniform. That question is answerable with existing data and existing tools, and the answer so far, to within  $\pm 2\%$ , is yes.

## Data and code availability

All figures are reproduced by the enumeration in Appendix A, which is exhaustive for  $n \leq 1009$  and therefore yields exact rationals rather than estimates. Datasets, the generator, and the control experiment are in the project repository at [repositoryURL].

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## A Reproduction code

The following is self-contained. For  $n \leq 1009$  it enumerates all  $(n-1)^3$  states, so the probabilities it returns are exact. Every assertion below is a restatement of a theorem in the body, and all of them pass.

```
import numpy as np, math

def enumerate_states(n):
    """Exhaustive sweep over the state space S of the structure lemma: (zeta, xi, u)."""
    v = np.arange(1, n, dtype=np.int64)
    Z, X = np.meshgrid(v, v, indexing='ij')
    Z, X = Z.ravel(), X.ravel()
    S = (Z + X) % n
    keep = S != 0
    Z, X, S = Z[keep], X[keep], S[keep]

    sig = Z + X # true integer sum
    nE = int(np.count_nonzero((sig % 2 == 0) &
                              (np.abs(Z - X) == sig // 2 + 1)))
    nE_line = int(np.count_nonzero((Z == 3*X + 2) | (X == 3*Z + 2))) # E structure thm
    assert nE == nE_line == 2 * ((n - 3) // 3) # exact E count

    q = a_ct = b_ct = 0
    for u in range(1, n):
        ZR = (Z * u) % n
        m = ZR > S
        z_, x_, s_, zr_ = Z[m], X[m], S[m], ZR[m]
        d = zr_ - s_
        a = s_ % d
        g = (a > 0) & (a != s_) & (a != zr_) # qualification
        z_, x_, s_, zr_, a = z_[g], x_[g], s_[g], zr_[g], a[g]
        q += a.size
        a_ct += int(np.count_nonzero(z_ == a)) # class A
        t = s_ + zr_
        both = (z_ % a == 0) & (t % x_ == 0) # class B
        if both.any():
            b_ct += int(np.count_nonzero(
                z_[both] // a[both] == t[both] // x_[both]))

    tot = (n - 1) ** 3
    assert abs(a_ct/tot - (q/tot)/(n-2)) < 1e-15 # exact A frequency
    return dict(n=n, PQ=q/tot, PA=a_ct/tot, PE=nE/(n-1)**2, PB=b_ct/tot)

for n in (101, 151, 251, 401, 503, 751, 1009):
    r = enumerate_states(n)
    print(f"n={r['n']:5d} Pr[Q]={r['PQ']:.6f} Pr[A]={r['PA']:.6e} "
          f"Pr[E]={r['PE']:.6e} Pr[B]*n^2/ln n={r['PB']*n*n/math.log(n):.3f}")

# --- secp256k1 -----
from decimal import Decimal as D, getcontext
getcontext().prec = 60
n = 0xFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFEBAAEDCE6AF48A03BBFD25E8CD0364141
PA = D(1) / 4 / (n - 2)
PE = D(2 * ((n - 3) // 3)) / D((n - 1) ** 2)
PB = D("0.19") * D(n).ln() / (D(n) * D(n))
print("Pr[A] = %.6E Pr[E] = %.6E Pr[B] = %.3E 1/n = %.6E"
      % (PA, PE, PB, D(1) / n))
```

## B Notation

|                                                                |                                             |
|----------------------------------------------------------------|---------------------------------------------|
| $n$                                                            | prime group order                           |
| $\zeta = s_{zk} = zk^{-1}, \tilde{\zeta} = s_{rxk} = rxk^{-1}$ | hidden components (Definition 1)            |
| $\sigma = \zeta + \tilde{\zeta} \in \{s, s + n\}$              | integer sum (2)                             |
| $s_{zr} = zr, d = s_{zr} - s, a = s \bmod d$                   | public auxiliaries (Definition 2)           |
| $m_1 = \zeta/a, m_2 = (s + s_{zr})/\tilde{\zeta}$              | divisibility ratios, taken in $\mathbb{Q}$  |
| $\mathcal{Q}$                                                  | qualification event (Definition 12)         |
| $C, \mathbb{E} C $                                             | candidate set of a rule; its expected size  |
| $\tau(\cdot), \gamma$                                          | divisor function; Euler–Mascheroni constant |

All modular arithmetic is modulo  $n$  unless a relation is explicitly stated to hold in  $\mathbb{Z}$ .